

Unit 3 Comprehensive Review

Directions: Show all work. No calculators.

A. Arithmetic Sequences and Modeling

1. Determine if the sequence is arithmetic. If it is, find the common difference. 11, 18, 25, 32, ...
2. Determine if the sequence is arithmetic. If it is, find the common difference. 3, 7, 12, 18, ...
3. Determine if the sequence is arithmetic. If it is, find the common difference. $-4, -9, -14, -19, \dots$
4. Find the common difference and write the next three terms: 30, 24, 18, 12, ...

5. Find the common difference and write the next three terms: $-15, -7, 1, 9, \dots$

6. Find a_{22} for the arithmetic sequence $5, 9, 13, 17, \dots$

7. Find a_{35} if $a_1 = 42$ and $d = -3$.

8. Find a_{17} if $a_1 = -11$ and $d = 9$.

9. Which term of the arithmetic sequence $9, 14, 19, 24, \dots$ is 104?

10. Write an explicit formula for the arithmetic sequence $-2, 5, 12, 19, \dots$

11. An arithmetic sequence has $a_1 = 16$ and $d = -5$. Write the first five terms and an explicit formula.

12. A set of bleachers has 26 seats in the first row. Each row behind it has 4 more seats than the row before. Write an explicit formula for the number of seats in row n , then find the number of seats in row 15.

13. A student saves \$30 during the first week and increases the amount saved by \$6 each week. Write an explicit formula for the amount saved during week n , then find the amount saved during week 14.

B. Functions, Function Notation, and Slope

14. If $f(x) = -4x + 3$, find $f(-5)$.

15. If $g(x) = x^2 + 2x - 7$, find $g(4)$.

16. If $h(x) = 5x - 8$, solve $h(x) = 27$.

17. If $p(x) = -3x + 10$, solve $p(x) = 1$.

18. Find the slope of the line through $(-7, 4)$ and $(1, 8)$.

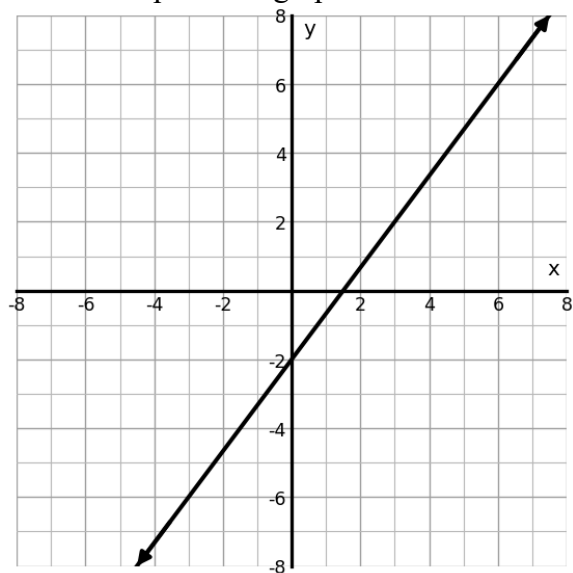
19. Find the slope of the line through $(2, -6)$ and $(7, 4)$.

20. Find the slope of the line through $(-3, 5)$ and $(5, -1)$.

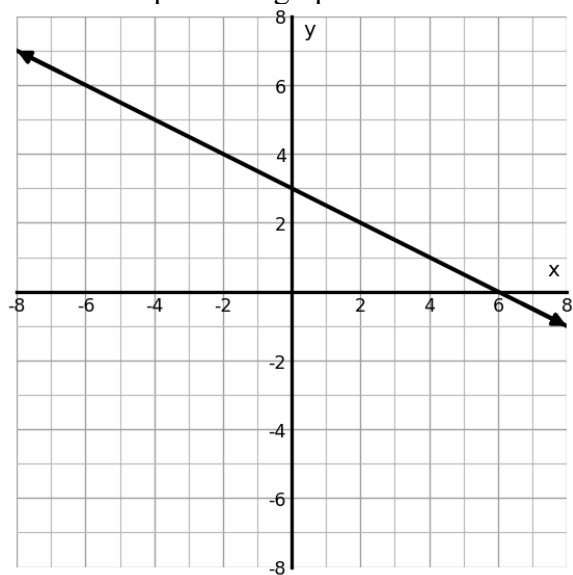
21. Are the given points collinear? Show how you determined your answer: $(-4, -5)$, $(0, 1)$, $(4, 7)$.

22. Are the given points collinear? Show how you determined your answer: $(-6, 4)$, $(-2, 0)$, $(2, -3)$.

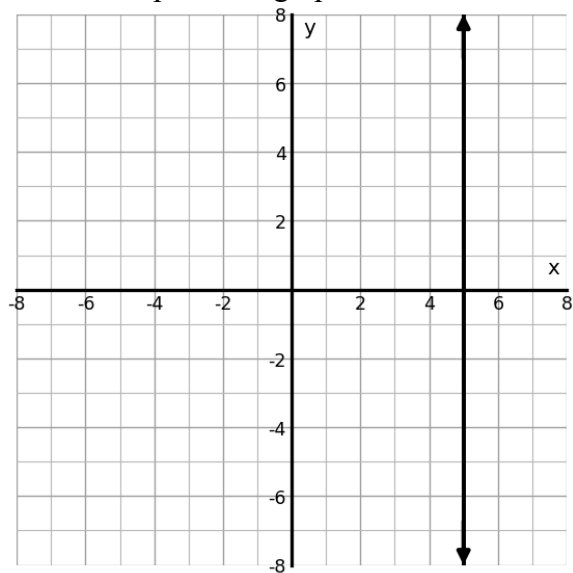
23. Find the slope of the graphed line.



24. Find the slope of the graphed line.



25. Find the slope of the graphed line.



C. Slope-Intercept Form and Point-Slope Form

26. Identify the slope and y-intercept of $y = -5x + 9$.

27. Identify the slope and y-intercept of $y = \frac{2}{3}x - 4$.

28. Write an equation in slope-intercept form with slope 3 and y-intercept -7 .

29. Write an equation in slope-intercept form with slope $-\frac{4}{5}$ and y-intercept 2.

30. Write the equation of the line through $(0, -3)$ and $(5, 7)$ in slope-intercept form.

31. Write the equation of the line through $(0, 8)$ and $(-4, 2)$ in slope-intercept form.

32. Write an equation of the line through $(-2, 3)$ and $(6, 7)$ in point-slope form.

33. Write an equation of the line through $(3, -4)$ and $(7, 6)$ in point-slope form.

34. Write an equation of the line through $(-4, 1)$ and $(2, 10)$ in point-slope form.

35. Write a point-slope equation for the line with slope -3 through $(-5, 2)$.

36. A line goes through the points $(-6, 8)$, $(-2, 2)$, $(2, -4)$, $(6, -10)$. Write 4 different point-slope equations for the line.

37. Convert $y - 3 = 4(x + 2)$ to slope-intercept form.

38. Convert $y + 5 = -\frac{3}{4}(x - 4)$ to slope-intercept form.

39. Convert $y - 6 = -\frac{5}{3}(x + 3)$ to slope-intercept form.

40. A phone plan charges a \$25 monthly fee plus \$8 per gigabyte of data. Write a linear function for the total cost after x gigabytes. Then find the total cost after 6 gigabytes.

41. A delivery service charges a \$9 starting fee plus \$2.25 per mile. Write a linear function for the total cost after x miles. Then find the total cost after 12 miles.

D. Recursive Formulas

42. Write a recursive formula to represent the sequence 12, 17, 22, 27,

43. Write a recursive formula to represent the sequence 31, 23, 15, 7,

44. Write the first 6 terms of a sequence if $a_1 = -6$, $a_n = a_{n-1} + 7$.

45. Write the first 6 terms of a sequence if $a_1 = 35$, $a_n = a_{n-1} - 6$.

46. A sequence has recursive formula $a_n = a_{n-1} + 9$ with $a_1 = -14$. What is the common difference for the sequence? Write the first 5 terms of the sequence.

47. A sequence has recursive formula $a_n = a_{n-1} - 4$ with $a_1 = 18$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
48. Convert $a_n = a_{n-1} + 6$ with $a_1 = 3$ to an explicit formula.
49. An arithmetic sequence has $a_1 = -8$ and $a_5 = 12$. Find the common difference and write a recursive formula for the sequence.
50. An arithmetic sequence has $a_1 = 14$ and $a_8 = -7$. Find the common difference and write a recursive formula for the sequence.